CSE 242 Assignment 3: SVMs, K-means, and K-NN

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Question 1: Hard-margin linear SVM

(a) Plot and support vectors

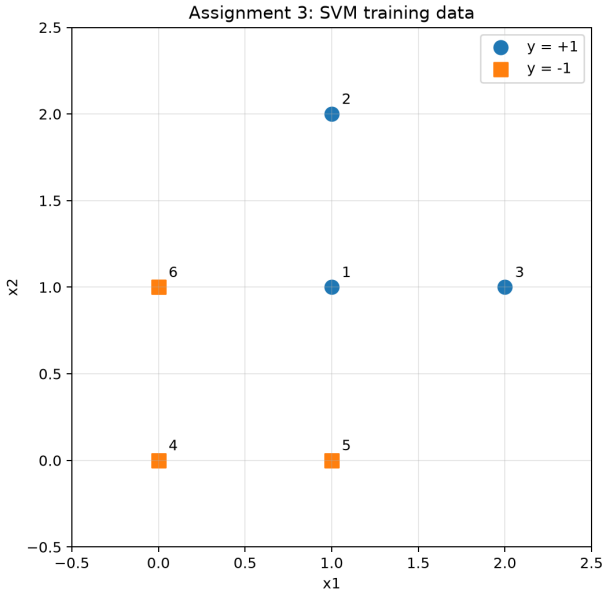


Figure 1: Given training data

The two classes are separated by a diagonal-looking gap, and the closest opposite-class pairs are $(1,1)$ with $(1,0)$ and $(1,1)$ with distance 1 each. Hence, the support vectors are:
Points 1 (1,1), 5 (1,0), 6 (0,1).

(b) Boundary and margin

each support vector satisfies $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$:

$$\begin{aligned}
&\text{point 1 } (1,1),\ y=+1: \quad w_1 + w_2 + b = 1,\\
&\text{point 5 } (1,0),\ y=-1: \quad w_1 + b = -1,\\
&\text{point 6 } (0,1),\ y=-1: \quad w_2 + b = -1.
\end{aligned}$$

Subtracting the last two equations gives $w_1 = w_2$ (say w). Then $2w + b = 1$ and $w + b = -1$, so $w = 2$ and $b = -3$:

$$\boxed{\mathbf{w} = (2,2), \quad b = -3, \quad 2x_1 + 2x_2 - 3 = 0}$$

Geometric margin:

$$\gamma = \frac{1}{\|\mathbf{w}\|} = \frac{1}{\sqrt{2^2 + 2^2}} = \frac{1}{2\sqrt{2}}$$

The full width between the two supporting boundaries ($\mathbf{w}^\top \mathbf{x} + b = \pm 1$) is: $2/\|\mathbf{w}\| = \sqrt{2}/2$

Question 2: K-means

(a) Comments: in code

(b) Results

Data: `make_blobs(n_samples=4000, centers=4, cluster_std=0.60, random_state=0)`, 100 iterations, seed 0. The true labels are used only for the reference plot.

k	Inertia (SSE)	Cluster sizes
2	15222.21	1017, 2983
3	6826.13	1008, 1992, 1000
4	2794.24	1000, 998, 1002, 1000
6	2396.79	1000, 299, 997, 407, 319, 978
10	1914.79	365, 350, 253, 261, 988, 997, 123, 213, 285, 165

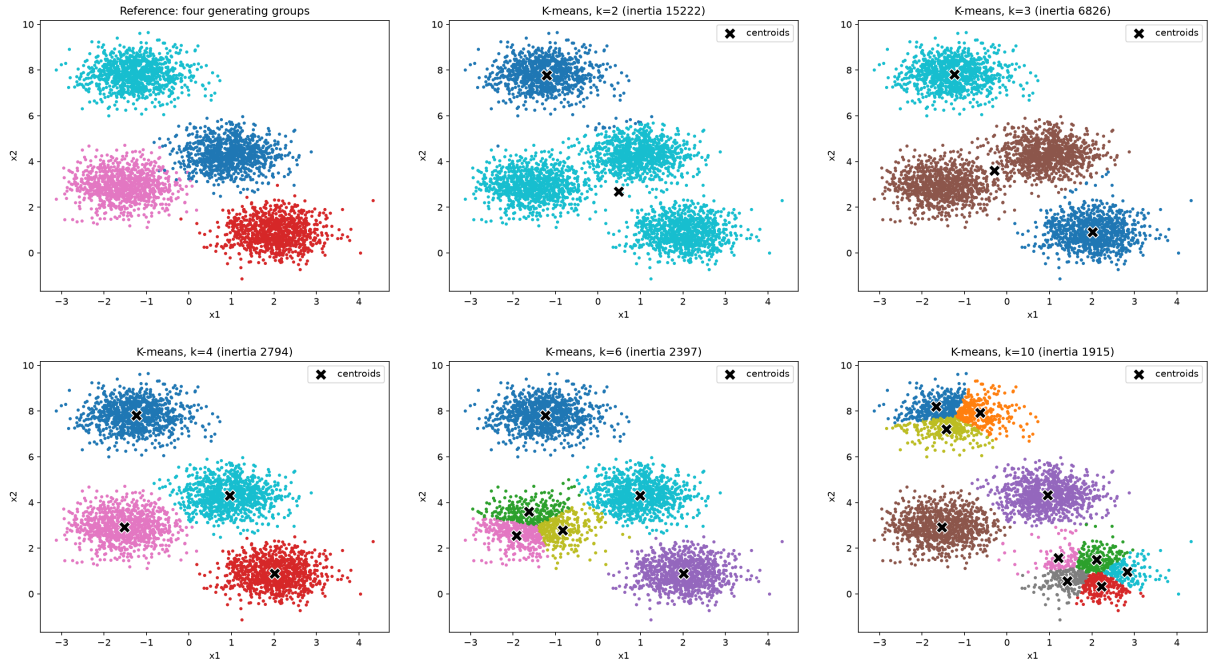


Figure 2: K-means for $k = 2, 3, 4, 6, 10$

Discussion: The data contain four groups of 1000 points each. With $k = 2, 3$ there are too few centroids, so some true groups are merged into one cluster and the error is huge. At $k = 4$ the clusters match the four generating groups almost exactly. For $k = 6, 10$, there are more centroids than groups, so some true groups are split into smaller ones. Inertia (sum of squared errors) always

decreases as k grows, so a lesser inertia value does not mean a better clustering always. zthe "elbow" at $k = 4$ matches the best.

Question 3: Noisy-label SVM and K-NN repair

Method	Mean test accuracy (20 trials)
Baseline (noisy labels)	0.8280
K-NN repaired labels	0.8644
Difference (K-NN − baseline)	+0.0364

Did K-NN repair help? Yes, mean accuracy improved by about 3.6%.

Why it can help. Label noise is independent of position. In regions dominated by one class, most of a point's 5 neighbours carry the correct label even though each individual label is wrong with probability 0.2–0.35, so the majority vote has a lower error rate than the labels.

Why it can hurt. The two classes overlap strongly, and the negative class is more spread out. Near the decision boundary, or in sparse regions, the neighbours come from both classes, so the votes can flip a correct label to a wrong one. Because positives are corrupted more often (0.35 vs. 0.20), the vote is also biased toward the negative class there. Here the benefits outweighed these effects.